

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398728
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Hirano; Takayuki et al.

Centrifugal compressor

Abstract

A connection passage is formed by a first impeller housing member, a motor housing member, and a second impeller housing member. This prevents, for example, the formation of a gap between a housing and a pipe, which is separate from the housing, in a centrifugal compressor like in a configuration in which a first discharge port and a second suction port are connected to each other by attaching the pipe to the housing. The connection passage is formed only by sequentially overlapping and coupling the first impeller housing member, the motor housing member, and the second impeller housing member with each other in an axial direction of the rotary shaft. This eliminates, for example, the need to provide a structure in the pipe that absorbs the dimensional tolerances that occur between the housing and the pipe when attaching the pipe to the housing.

Inventors:	Hirano; Takayuki (Kariya, JP), Suzuki; Junya (Kariya, JP)
Applicant:	KABUSHIKI KAISHA TOYOTA JIDOSHOKKI (Kariya, JP)
Family ID:	1000008781161
Assignee:	KABUSHIKI KAISHA TOYOTA JIDOSHOKKI (Kariya, JP)
Appl. No.:	18/953739
Filed:	November 20, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250180035 A1	Jun. 05, 2025

Foreign Application Priority Data

JP	2023-203146	Nov. 30, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: F04D29/42 (20060101); F04D17/10 (20060101); F04D25/06 (20060101); F04D29/58 (20060101)

U.S. Cl.:

CPC F04D29/4206 (20130101); F04D17/10 (20130101); F04D25/06 (20130101); F04D29/582 (20130101);

Field of Classification Search

CPC: F04D (29/4206); F04D (17/10); F04D (25/06); F04D (29/582)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11274679	12/2021	Sun	N/A	F04D 29/4206
2015/0167689	12/2014	Oda	415/110	F04D 1/04
2015/0233384	12/2014	Thompson et al.	N/A	N/A
2015/0275920	12/2014	Thompson et al.	N/A	N/A
2015/0308456	12/2014	Thompson et al.	N/A	N/A
2017/0227012	12/2016	Hirata	N/A	F04D 29/52

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2015-209845	12/2014	JP	N/A

Primary Examiner: Nguyen; Hoang M

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

BACKGROUND

1. Field

(1) The present disclosure relates to a centrifugal compressor.

2. Description of Related Art

(2) The centrifugal compressor includes a rotary shaft, a motor, and a housing. The motor rotates the rotary shaft. Additionally, the centrifugal compressor may include a first impeller and a second impeller. The first impeller rotates integrally with the rotary shaft to compress a fluid. The second impeller may be located on the opposite side of the first impeller, with the motor located between the second impeller and the rotary shaft.

(3) The housing includes a first impeller chamber, a second impeller chamber, a first suction port, a first discharge port, a second suction port, and a second discharge port. The first impeller chamber accommodates the first impeller. The second impeller chamber accommodates the second impeller. The first suction port draws a fluid into the first impeller chamber. The first discharge port discharges the fluid compressed by the rotation of the first impeller. The second suction port draws a fluid into the second impeller chamber. The centrifugal compressor includes a connection

passage. The second discharge port discharges the fluid compressed by the rotation of the second impeller. The connection passage connects the first discharge port to the second suction port such that a fluid flows from the first discharge port toward the second suction port. The connection passage is formed by a pipe that is separate from the housing as described in, for example, Japanese Laid-Open Patent Publication No. 2015-209845. The pipe is attached to the housing to connect the first discharge port to the second suction port through the connection passage.

(4) Japanese Laid-Open Patent Publication No. 2015-209845 discloses a configuration in which the pipe, which is separate from the housing, is attached to the housing to connect the first discharge port to the second suction port through the connection passage. In this configuration, the centrifugal compressor includes a gap between the housing and the pipe. The gap between the housing and the pipe is a dead space in the centrifugal compressor. Thus, the centrifugal compressor is unnecessarily enlarged by an amount corresponding to the gap between the housing and the pipe.

(5) Moreover, dimensional tolerances occur between the housing and the pipe. This necessitates, for example, providing a structure in the pipe that can absorb the dimensional tolerances that occur between the housing and the pipe when attaching the pipe to the housing, resulting in a factor that reduces the production efficiency of the centrifugal compressor. Accordingly, it is desirable to improve the production efficiency of the centrifugal compressor while also limiting its increase in size.

SUMMARY

(6) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(7) An aspect of the present disclosure provides a centrifugal compressor. The centrifugal compressor includes a rotary shaft, a motor that rotates the rotary shaft, a first impeller that rotates integrally with the rotary shaft to compress a fluid, and a second impeller located on an opposite side of the rotary shaft from the first impeller. The motor is arranged between the first impeller and the second impeller. The second impeller rotates integrally with the rotary shaft. The centrifugal compressor also includes a housing. The housing includes a first impeller housing member, a motor housing member, a second impeller housing member. The first impeller housing member defines a first impeller chamber accommodating the first impeller. The first impeller housing member includes a first suction port that draws the fluid into the first impeller chamber and a first discharge port that discharges the fluid compressed by rotation of the first impeller. The motor housing member defines a motor chamber accommodating the motor. The second impeller housing member defines a second impeller chamber accommodating the second impeller. The second impeller housing member includes a second suction port that draws the fluid into the second impeller chamber and a second discharge port that discharges the fluid compressed by rotation of the second impeller. The centrifugal compressor further includes a connection passage that connects the first discharge port to the second suction port such that the fluid flows from the first discharge port toward the second suction port. The connection passage includes a first impeller housing member connection passage located in the first impeller housing member and connected to the first discharge port, a second impeller housing member connection passage located in the second impeller housing member and connected to the second suction port, and a motor housing member connection passage extending through the motor housing member and connecting the first impeller housing member connection passage to the second impeller housing member connection passage. The connection passage is formed by sequentially overlapping the first impeller housing member, the motor housing member, and the second impeller housing member with each other in an axial direction of the rotary shaft.

(8) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-sectional view of a centrifugal compressor.
- (2) FIG. 2 is a cross-sectional view of the first impeller housing member.
- (3) FIG. 3 is a cross-sectional view of the second impeller housing member.
- (4) FIG. 4 is a rear view of the second impeller housing member base.
- (5) FIG. 5 is a rear view of the second impeller housing member base and the second impeller housing member cover.
- (6) Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

- (7) This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.
- (8) Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.
- (9) In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”
- (10) A centrifugal compressor according to an embodiment will now be described with reference to FIGS. 1 to 5. The centrifugal compressor of the present embodiment is mounted on a fuel cell electric vehicle. The centrifugal compressor compresses air as fluid.
- (11) Basic Structure of Centrifugal Compressor
- (12) As shown in FIG. 1, a centrifugal compressor **100** includes a housing **10**. The housing **10** is made of metal. The housing **10** is made of, for example, aluminum. The housing **10** includes a motor housing member **10a**, a first impeller housing member **10b**, and a second impeller housing member **10c**. The motor housing member **10a** includes a first motor housing member **14** and a connection passage defining housing member **13**. The first impeller housing member **10b** includes a first compressor housing member **11** and a second plate **17**. The second impeller housing member **10c** includes a second compressor housing member **12** and a third plate **18**. The second compressor housing member **12** includes a second impeller housing member base **12a** and a second impeller housing member cover **19**. In other words, the second impeller housing member **10c** includes the second impeller housing member base **12a** and the second impeller housing member cover **19**.
- (13) The first motor housing member **14** includes an end wall **14a** and a circumferential wall **14b**. The first motor housing member **14** has a cylindrical shape. The end wall **14a** has a plate shape. The circumferential wall **14b** extends in a tubular shape from an outer circumferential portion of the end wall **14a**. The first motor housing member **14** is manufactured through die casting.
- (14) The connection passage defining housing member **13** includes a second motor housing member **15** and a first plate **16**. In other words, the motor housing member **10a** includes the first motor housing member **14** and the second motor housing member **15**. The second motor housing member **15** and the first plate **16** are manufactured through die casting.
- (15) The second motor housing member **15** has a tubular shape. The second motor housing member **15** includes a motor housing member accommodation hole **15c**. The second motor housing member **15** includes an accommodation hole defining surface **15a**. The accommodation hole defining

surface **15a** defines the motor housing member accommodation hole **15c**. The motor housing member accommodation hole **15c** extends in the axial direction of the second motor housing member **15** and opens in the opposite end faces of the second motor housing member **15** in the axial direction. The motor housing member accommodation hole **15c** accommodates the first motor housing member **14** such that the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14** faces the accommodation hole defining surface **15a** in the radial direction of the second motor housing member **15**. The axis of the second motor housing member **15** coincides with the axis of the circumferential wall **14b** of the first motor housing member **14**.

(16) The centrifugal compressor **100** includes a cooling medium passage **13a**. The cooling medium passage **13a** is defined by the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14** and the accommodation hole defining surface **15a**. The cooling medium passage **13a** is defined such that coolant flows between the first motor housing member **14** and the second motor housing member **15**. In other words, the cooling medium passage **13a** is defined by the first motor housing member **14** and the second motor housing member **15**. The cooling medium passage **13a** extends along the circumferential wall **14b** of the first motor housing member **14** and has an annular shape surrounding the circumferential wall **14b**. The second motor housing member **15** has a supply port **13e** and a discharge port **13f**. The supply port **13e** supplies coolant to the cooling medium passage **13a**. The discharge port **13f** discharges, to the outside of the housing **10**, the coolant that has flowed through the cooling medium passage **13a**. The coolant flowing through the cooling medium passage **13a** cools the first motor housing member **14** and the second motor housing member **15**.

(17) Fins **14d** are provided on the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14**. The fins **14d** protrude from the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14** into the cooling medium passage **13a**. The fins **14d** are annular thin plates that extend along the outer circumferential surface of the circumferential wall **14b**. The fins **14d** are arranged at intervals in the axial direction of the first motor housing member **14** on the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14**. The fins **14d** are formed integrally with the first motor housing member **14**. Coolant flows through the cooling medium passage **13a**, along each fin **14d**, in the circumferential direction of the first motor housing member **14** and the second motor housing member **15**.

(18) The first plate **16** is coupled to the end of the circumferential wall **14b** of the first motor housing member **14** on a side opposite to the end wall **14a** and to a first end of the second motor housing member **15**. The first plate **16** is coupled to the circumferential wall **14b** of the first motor housing member **14** and the second motor housing member **15**, with the thickness direction of the first plate **16** coinciding with the axial direction of the circumferential wall **14b** of the first motor housing member **14** and the axial direction of the second motor housing member **15**. The first plate **16** closes the opening of the circumferential wall **14b** of the first motor housing member **14**. Additionally, the first motor housing member **14** and the first plate **16** define a motor chamber **14c**. Thus, the motor housing member **10a** defines the motor chamber **14c**.

(19) The first plate **16** includes a first bearing retainer **21**. The first bearing retainer **21** is cylindrical. The first bearing retainer **21** protrudes into the motor chamber **14c**. The axis of the first bearing retainer **21** coincides with the axis of the circumferential wall **14b** of the first motor housing member **14**.

(20) The end wall **14a** of the first motor housing member **14** includes a second bearing retainer **25**. The second bearing retainer **25** is cylindrical. The second bearing retainer **25** protrudes into the motor chamber **14c**. The axis of the second bearing retainer **25** coincides with the axis of the first bearing retainer **21**.

(21) The second plate **17** is manufactured through die casting. The second plate **17** is coupled to the end face of the first plate **16** on a side opposite to the first motor housing member **14** and the

second motor housing member **15**. The second plate **17** is coupled to the first plate **16**, with the thickness direction of the second plate **17** coinciding with the thickness direction of the first plate **16**. The second plate **17** has a first insertion hole **23**. The axis of the first insertion hole **23** coincides with the axis of the first bearing retainer **21**. The first insertion hole **23** is connected to the interior of the first bearing retainer **21**.

(22) The third plate **18** is manufactured through die casting. The third plate **18** is coupled to the outer surface of the end wall **14a** of the first motor housing member **14** and to a second end of the second motor housing member **15**. The third plate **18** is coupled to the end wall **14a** of the first motor housing member **14** and the second motor housing member **15**, with the thickness direction of the third plate **18** coinciding with the thickness direction of the end wall **14a** of the first motor housing member **14**.

(23) The third plate **18** has a second insertion hole **26**. The second insertion hole **26** extends through the third plate **18** in the thickness direction of the third plate **18**. The second insertion hole **26** is connected to the interior of the second bearing retainer **25**. The axis of the second insertion hole **26** coincides with the axis of the second bearing retainer **25**.

(24) The centrifugal compressor **100** includes a motor **20**. The motor **20** is accommodated in the motor chamber **14c**. Thus, the motor chamber **14c** accommodates the motor **20**. The first motor housing member **14** surrounds the motor **20**. The heat generated from the motor **20** is dissipated to the first motor housing member **14**. The first motor housing member **14** is cooled by the coolant flowing through the cooling medium passage **13a**. Thus, the heat generated by the motor **20** is efficiently dissipated to the first motor housing member **14**. As a result, the motor **20** is cooled. That is, the motor housing member **10a** includes the cooling medium passage **13a**, through which coolant as cooling medium flows to cool the motor **20**.

(25) The first compressor housing member **11** is manufactured through die-casting. The first compressor housing member **11** has a circular first suction port **35** through which air is drawn in. Thus, the first impeller housing member **10b** includes the first suction port **35**. The first compressor housing member **11** is coupled to the end face of the second plate **17** on a side opposite to the first plate **16**, with the axis of the first suction port **35** coinciding with the axis of the first insertion hole **23**. The first suction port **35** opens in the end face of the first compressor housing member **11** on a side opposite to the second plate **17**. Air that has been cleaned by an air cleaner (not shown) flows through the first suction port **35**.

(26) The centrifugal compressor **100** includes a first impeller chamber **36**, a first discharge chamber **37**, and a first diffuser passage **38**. The first impeller chamber **36**, the first discharge chamber **37**, and the first diffuser passage **38** are located between the first compressor housing member **11** and the second plate **17**. Thus, the first impeller housing member **10b** defines the first impeller chamber **36**. The first impeller chamber **36** is connected to the first suction port **35**. The first suction port **35** draws air into the first impeller chamber **36**. The first discharge chamber **37** is located around the first impeller chamber **36** and extends about the axis of the first suction port **35**. The first diffuser passage **38** connects between the first impeller chamber **36** and the first discharge chamber **37**. The first impeller chamber **36** is connected to the first insertion hole **23**.

(27) As shown in FIG. 2, the first compressor housing member **11** includes a first discharge port **39**. Thus, the first impeller housing member **10b** includes the first discharge port **39**. A first end of the first discharge port **39** is connected to the first discharge chamber **37**.

(28) Referring to FIG. 1, the second impeller housing member base **12a** is manufactured through die casting. The second impeller housing member base **12a** has a circular second suction port **40**. The second suction port **40** opens in the second impeller housing member base **12a**. Thus, the second impeller housing member **10c** includes the second suction port **40**. The second impeller housing member base **12a** is coupled to the end face of the third plate **18** on a side opposite to the first motor housing member **14** and the second motor housing member **15**, with the axis of the second suction port **40** coinciding with the axis of the second insertion hole **26**.

(29) The centrifugal compressor **100** includes a second impeller chamber **41**, a second discharge chamber **42**, and a second diffuser passage **43**. The second impeller chamber **41**, the second discharge chamber **42**, and the second diffuser passage **43** are located between the second impeller housing member base **12a** and the third plate **18**. Thus, the second impeller housing member **10c** defines the second impeller chamber **41**. The second impeller chamber **41** is connected to the second suction port **40**. The second suction port **40** draws air into the second impeller chamber **41**. The second discharge chamber **42** is located around the second impeller chamber **41** and extends about the axis of the second suction port **40**. The second diffuser passage **43** connects the second impeller chamber **41** between the second discharge chamber **42**. The second impeller chamber **41** is connected to the second insertion hole **26**.

(30) The second impeller housing member base **12a** has a second discharge port **44**. The second discharge port **44** opens in the second impeller housing member base **12a**. In other words, the second impeller housing member **10c** has the second discharge port **44**. A first end of the second discharge port **44** is connected to the second discharge chamber **42**. A second end of the second discharge port **44** opens in the outer circumferential surface of the second impeller housing member base **12a**.

(31) A supply pipe **45** is connected to the second discharge port **44**. The supply pipe **45** is connected to a fuel cell stack **46**. A first end of the supply pipe **45** is connected to the second discharge port **44**. A second end of the supply pipe **45** is connected to the fuel cell stack **46**.

(32) The centrifugal compressor **100** includes a rotary shaft **50**. The rotary shaft **50**, with its axis coinciding with the axis of the circumferential wall **14b** of the first motor housing member **14**, extends across the motor chamber **14c**. A first end, which is one end in the axial direction, of the rotary shaft **50** protrudes into the first impeller chamber **36** from the inside of the motor chamber **14c**, through the interior of the first bearing retainer **21** and the first insertion hole **23**. A second end, which is the other end in the axial direction, of the rotary shaft **50** protrudes into the second impeller chamber **41** from the interior of the motor chamber **14c**, through the interior of the second bearing retainer **25** and the second insertion hole **26**.

(33) The motor **20** includes a cylindrical stator **52** and a rotor **51**. The rotor **51** is fixed to the rotary shaft **50**. The rotor **51** includes a cylindrical rotor core **53** that is fixed to the rotary shaft **50** and permanent magnets (not shown) that are provided in the rotor core **53**. The rotor **51** is installed on the rotary shaft **50** and rotates integrally with the rotary shaft **50**.

(34) The stator **52** is fixed to the first motor housing member **14**. The stator **52** is disposed on the outer side of the rotor **51**. The stator **52** includes a cylindrical stator core **54** and a coil **55**. The stator core **54** is fixed to the inner circumferential surface of the circumferential wall **14b** of the first motor housing member **14**. In other words, the motor housing member **10a** includes the first motor housing member **14** having an inner circumferential surface to which the stator **52** is fixed. The coil **55** is wound around the stator core **54**. The rotary shaft **50** rotates integrally with the rotor **51** when current flows through the coils **55** from a battery (not shown). Therefore, the motor **20** rotates the rotary shaft **50**.

(35) The centrifugal compressor **100** includes a first dynamic plain bearing **56** and a second dynamic plain bearing **57**. The first dynamic plain bearing **56** is cylindrical. The first dynamic plain bearing **56** is retained by the first bearing retainer **21**. The first dynamic plain bearing **56** supports the rotary shaft **50** to be rotatable with respect to the first plate **16**.

(36) The second dynamic plain bearing **57** is cylindrical. The second dynamic plain bearing **57** is retained by the second bearing retainer **25**. The second dynamic plain bearing **57** supports the rotary shaft **50** to be rotatable with respect to the end wall **14a** of the first motor housing member **14**. Thus, the first dynamic plain bearing **56** and the second dynamic plain bearing **57** support the rotary shaft **50** to be rotatable with respect to the housing **10**.

(37) The first dynamic plain bearing **56** and the second dynamic plain bearing **57** support the rotary shaft **50** to be rotatable in the radial direction with respect to the housing **10**.

(38) The centrifugal compressor **100** may include a known thrust bearing that supports the rotary shaft **50** to be rotatable in a thrust direction with respect to the housing **10**.

(39) The centrifugal compressor **100** includes a first impeller **61** and a second impeller **62**. The first impeller **61** is coupled to the first end of the rotary shaft **50**. The first impeller **61** is accommodated in the first impeller chamber **36**. The first impeller chamber **36** thus accommodates the first impeller **61**. The first impeller **61** rotates integrally with the rotary shaft **50** to compress air.

(40) The second impeller **62** is coupled to the second end of the rotary shaft **50**. In other words, the second impeller **62** is located on the opposite side of the first impeller **61**, with the motor **20** located between the second impeller **62** and the rotary shaft **50**. The second impeller **62** is accommodated in the second impeller chamber **41**. The second impeller chamber **41** thus accommodates the second impeller **62**. The second impeller **62** rotates integrally with the rotary shaft **50** to compress the air that has been compressed by the rotation of the first impeller **61**. The first impeller **61** and the second impeller **62** rotate to compress air that is supplied to the fuel cell stack **46**.

(41) The first impeller **61** and the second impeller **62** rotate integrally with the rotary shaft **50** through the driving of the motor **20**. Thus, the motor **20** rotates the first impeller **61** and the second impeller **62**.

(42) Inverter

(43) The centrifugal compressor **100** includes an inverter **65**. The inverter **65** is electrically connected to the motor **20**. The inverter **65** controls the driving of the motor **20** by controlling power that is supplied to the coil **55**.

(44) The centrifugal compressor **100** includes an inverter housing member **66**. The inverter housing member **66** accommodates the inverter **65**. The inverter housing member **66** is provided on the outer surface of the second motor housing member **15**. In other words, the inverter **65** is provided on a housing outer surface **10e**, which is the outer surface of the housing **10**. The heat generated from the inverter **65** is dissipated through the inverter housing member **66** to the second motor housing member **15**. Since the second motor housing member **15** is cooled by the coolant flowing through the cooling medium passage **13a**, the heat generated from the inverter **65** is efficiently dissipated to the second motor housing member **15**. As a result, the inverter **65** is cooled.

(45) The centrifugal compressor **100** includes a seal member **63**. The seal member **63** is located between the first insertion hole **23** and the rotary shaft **50**. The seal member **63** restricts air leakage toward the interior of the motor chamber **14c** from the first impeller chamber **36** through the first insertion hole **23** and the interior of the first bearing retainer **21**. The seal member **63** is, for example, a seal ring.

(46) The centrifugal compressor **100** includes a seal member **64**. The seal member **64** is located between the second insertion hole **26** and the rotary shaft **50**. The seal member **64** restricts air leakage toward the interior of the motor chamber **14c** from the second impeller chamber **41** through the second insertion hole **26** and the interior of the second bearing retainer **25**. The seal member **64** is, for example, a seal ring.

(47) Connection Passage

(48) As shown in FIGS. **1** and **2**, the centrifugal compressor **100** includes a connection passage **70**. The connection passage **70** connects the first discharge port **39** to the second suction port **40** such that air flows from the first discharge port **39** toward the second suction port **40**. The connection passage **70** includes a first impeller housing member connection passage **101**, a second impeller housing member connection passage **102**, and a motor housing member connection passage **13b**. Thus, in the present embodiment, the connection passage **70** is defined by the first impeller housing member connection passage **101**, the second impeller housing member connection passage **102**, and the motor housing member connection passage **13b**.

(49) The first compressor housing member **11** includes a first passage **11a**. The first compressor housing member **11** has a first passage defining surface **11b**. The first passage defining surface **11b** defines the first passage **11a**. A first end of the first passage **11a** is connected to the first discharge

port **39**. A second end of the first passage **11a** opens in the end face of the first compressor housing member **11** located toward the second plate **17** and at a portion located outward from the first discharge chamber **37** in the radial direction of the rotary shaft **50**. The first passage defining surface **11b** is a curved surface extending from an opening edge at the second end of the first passage **11a**.

(50) The second plate **17** has a second plate hole **17a**. The second plate **17** includes a second plate hole defining surface **17b**. The second plate hole defining surface **17b** defines the second plate hole **17a**. The second plate hole defining surface **17b** is continuous with the first passage defining surface **11b**. In other words, the second plate hole **17a** is connected to the first passage **11a**. The second plate hole **17a** opens in the opposite end faces of the second plate **17** in the thickness direction. The second plate hole **17a** is connected to the second end of the first passage **11a**.

(51) The first impeller housing member connection passage **101** is defined by the first passage **11a** and the second plate hole **17a**. In other words, the first impeller housing member connection passage **101** is formed in the first impeller housing member **10b**. Thus, the first impeller housing member connection passage **101** is formed in the housing **10**. A first end of the first impeller housing member connection passage **101** is connected to the first discharge port **39**. Air flows through the first impeller housing member connection passage **101** from the first discharge port **39**. Thus, the first impeller housing member connection passage **101** is located in the first impeller housing member **10b** and connected to the first discharge port **39**. A second end of the first impeller housing member connection passage **101** opens in the end face of the second plate **17** located toward the first plate **16** and at a portion located outward from the first discharge chamber **37** in the radial direction of the rotary shaft **50**.

(52) A boundary **L1** between the first end of the first impeller housing member connection passage **101** and the first discharge port **39** is indicated by the long dashed double-short dashed line in FIG. 2. The portion of the first impeller housing member **10b** corresponding to the boundary **L1** between the first end of the first impeller housing member connection passage **101** and the first discharge port **39** corresponds to the location where a first end of a pipe that has been conventionally used to connect the first discharge port **39** to the second suction port **40** is connected. The first impeller housing member connection passage **101** of the present embodiment extends from a portion corresponding to the location where the first end of the conventional pipe is connected.

(53) The second compressor housing member **12** includes a second passage **12b**. As shown in FIGS. 1, 3, and 4, the second passage **12b** includes a first axial passage **401**, a second axial passage **402**, and a radial passage **403**. The first axial passage **401** is formed in the second impeller housing member base **12a**. In other words, the first axial passage **401**, which is a part of the second passage **12b**, is formed in the second impeller housing member base **12a**. The second impeller housing member base **12a** includes a first axial passage defining surface **401a**, which defines the first axial passage **401**. The first axial passage **401** is located outward from the second suction port **40** in the radial direction of the rotary shaft **50** and extends in the axial direction of the rotary shaft **50**. The first axial passage **401** extends through the second impeller housing member base **12a** in the axial direction of the rotary shaft **50**. A first end of the first axial passage **401** opens in the end face of the second impeller housing member base **12a** located toward the third plate **18**. A second end of the first axial passage **401** opens in the end face of the second impeller housing member base **12a** on a side opposite to the third plate **18**. The end face of the second impeller housing member base **12a** on the side opposite to the third plate **18** is an axial end face **12d**, which is an end face of the second impeller housing member base **12a** located in the axial direction of the rotary shaft **50**. Thus, the first axial passage **401** opens in the axial end face **12d**.

(54) The first axial passage **401** includes a third suction port **401b** that is an opening at the first end and a third discharge port **401c** that is an opening at the second end. In other words, the second impeller housing member **10c** includes the third suction port **401b** and the third discharge port **401c**. The third suction port **401b** and the third discharge port **401c** open in the second impeller

housing member base **12a**.

(55) The second axial passage **402** is formed in the second impeller housing member base **12a**. A first end of the second axial passage **402** is connected to the second suction port **40**. In other words, the second passage **12b** is located in the second compressor housing member **12** and connected to the second suction port **40**. A second end of the second axial passage **402** opens in the axial end face **12d**. Thus, the second suction port **40** extends from the second impeller chamber **41** in the axial direction of the rotary shaft **50** and opens in the axial end face **12d** through the second axial passage **402**. The axis of the second suction port **40** coincides with the axis of the second axial passage **402**. The second impeller housing member base **12a** includes a second axial passage defining surface **402a**, which defines the second axial passage **402**.

(56) Referring to FIGS. **1** and **5**, the second impeller housing member cover **19** is manufactured through die casting. The second impeller housing member cover **19** is coupled to the axial end face **12d** of the second impeller housing member base **12a**. The second impeller housing member cover **19** seals the opening of the second suction port **40** located toward the axial end face **12d** and the opening of the first axial passage **401** located toward the axial end face **12d**.

(57) As shown in FIG. **1**, the second impeller housing member cover **19** includes a facing surface **19b** that faces the axial end face **12d** of the second impeller housing member base **12a**. In other words, the second impeller housing member cover **19** includes the facing surface **19b**, which faces the second impeller housing member base **12a** in the axial direction of the rotary shaft **50**. The facing surface **19b** is flat.

(58) The facing surface **19b** of the second impeller housing member cover **19** includes a guide **19d**. The guide **19d** has the shape of a groove. Thus, the second impeller housing member **10c** includes the second impeller housing member cover **19** with the groove-shaped guide **19d**. With the second impeller housing member cover **19** coupled to the axial end face **12d** of the second impeller housing member base **12a**, the interior of the guide **19d** connects the second end of the first axial passage **401** to the second end of the second axial passage **402**. In other words, the guide **19d** is connected to the first axial passage **401** through the third discharge port **401c**. The section between an inner surface **19c** of the guide **19d** and the axial end face **12d** of the second impeller housing member base **12a** is the radial passage **403**. Thus, the section between the second impeller housing member cover **19** and the second impeller housing member base **12a** includes the radial passage **403**, which is a part of the second impeller housing member connection passage **102**, which extends in the radial direction of the rotary shaft **50**, and through which air from the first axial passage **401** flows toward the second suction port **40**. In this manner, the guide **19d** defines the radial passage **403**.

(59) The inner surface **19c** of the guide **19d** includes a first cover connection surface **191**, a second cover connection surface **192**, and a bottom surface **193**. The bottom surface **193** is a flat surface extending in the radial direction of the rotary shaft **50**. The first cover connection surface **191** extends from an opening edge **19e** of the guide **19d** and is connected to the bottom surface **193**. The first cover connection surface **191** extends from a portion of the opening edge **19e** of the guide **19d** located toward the first axial passage **401**. The first cover connection surface **191** is curved such that the direction extending from the opening edge **19e** of the guide **19d** toward the bottom surface **193** is oriented from the axial direction of the first axial passage **401** toward a direction orthogonal to the axial direction of the first axial passage **401**. The first cover connection surface **191** is curved such that the direction of the flow of air that has passed through the first axial passage **401** changes from the extending direction of the first axial passage **401** to the extending direction of the radial passage **403**. The first cover connection surface **191** smoothly connects the first axial passage defining surface **401a** to the bottom surface **193**.

(60) The second cover connection surface **192** extends from the opening edge **19e** of the guide **19d** and is connected to the bottom surface **193**. The second cover connection surface **192** extends from a portion of the opening edge **19e** of the guide **19d** located toward the second suction port **40**. The

second cover connection surface **192** is curved such that the direction extending from the opening edge **19e** of the guide **19d** toward the bottom surface **193** is oriented from the axial direction of the second axial passage **402** toward a direction orthogonal to the axial direction of the second axial passage **402**. The second cover connection surface **192** is curved such that the direction of the flow of air that has passed through the radial passage **403** is the direction in which the air flows into the second suction port **40**. The second cover connection surface **192** smoothly connects the second axial passage defining surface **402a** to the bottom surface **193**.

(61) The third plate **18** has a third plate hole **18a**. The third plate **18** includes a third plate hole defining surface **18b**. The third plate hole defining surface **18b** defines the third plate hole **18a**. The diameter of the third plate hole **18a** is the same as the diameter of the first axial passage **401**. The axis of the third plate hole **18a** coincides with the axis of the first axial passage **401**. The third plate hole defining surface **18b** is continuous with the first axial passage defining surface **401a**. The third plate hole **18a** opens in the opposite end faces of the third plate **18** in the thickness direction. The third plate hole **18a** is connected to the first end of the first axial passage **401** and connected to the third suction port **401b**.

(62) The second impeller housing member connection passage **102** is defined by the second passage **12b** and the third plate hole **18a**. In other words, the second impeller housing member **10c** defines the second impeller housing member connection passage **102**, which causes air to flow toward the second suction port **40**.

(63) A boundary **L2** between the second impeller housing member connection passage **102** and the second suction port **40** is shown by the long dashed double-short dashed line in FIG. **1**. The portion of the second impeller housing member **10c** corresponding to the boundary **L2** between the second impeller housing member connection passage **102** and the first suction port **35** corresponds to the location where a second end of the pipe, which has been conventionally used to connect the first discharge port **39** to the second suction port **40**, is connected. The second impeller housing member connection passage **102** of the present embodiment extends from a portion corresponding to the location where the second end of the conventional pipe is connected.

(64) The second motor housing member **15** has a connection hole **15d**. The second motor housing member **15** includes a connection hole defining surface **15b**. The connection hole defining surface **15b** defines the connection hole **15d**. The connection hole **15d** extends in the axial direction of the second motor housing member **15** and opens in the opposite end faces of the second motor housing member **15** in the axial direction. The connection hole **15d** is located outward from the motor housing member accommodation hole **15c** in the radial direction of the rotary shaft **50**. The extending direction of the connection hole **15d** coincides with the axial direction of the rotary shaft **50**. The axis of the connection hole **15d** coincides with the axis of the third plate hole **18a**. The diameter of the connection hole **15d** is the same as the diameter of the third plate hole **18a**. The connection hole defining surface **15b** is continuous with the third plate hole defining surface **18b**. The connection hole **15d** is connected to the third plate hole **18a**.

(65) The first plate **16** has a first plate hole **16a**. The first plate **16** includes a first plate hole defining surface **16b**. The first plate hole defining surface **16b** defines the first plate hole **16a**. The diameter of the first plate hole **16a** is equal to the diameters of the connection hole **15d** and the second plate hole **17a**. The axis of the first plate hole **16a** coincides with the axes of the connection hole **15d** and the second plate hole **17a**. The first plate hole defining surface **16b** is continuous with the connection hole defining surface **15b** and the second plate hole defining surface **17b**. The first plate hole **16a** opens in the opposite end faces of the first plate **16** in the thickness direction. A first end of the first plate hole **16a** is connected to one of the ends of the second plate hole **17a** in the axial direction that is not connected to the first passage **11a**. A second end of the first plate hole **16a** is connected to the first end of the connection hole **15d**.

(66) The connection hole **15d** and the first plate hole **16a** form the motor housing member connection passage **13b**. The connection hole defining surface **15b** and the first plate hole defining

surface **16b** form the third passage defining surface **13c**, which defines the motor housing member connection passage **13b**. Thus, the connection passage defining housing member **13** defines the motor housing member connection passage **13b**. The motor housing member connection passage **13b** extends through the motor housing member **10a** and connects the first impeller housing member connection passage **101** to the second impeller housing member connection passage **102**. The motor housing member connection passage **13b** extends through the second motor housing member **15**. The first end of the third plate hole **18a** is connected to the motor housing member connection passage **13b**. The third suction port **401b** faces the motor housing member connection passage **13b**, drawing in air in the axial direction of the rotary shaft **50**. The cooling medium passage **13a** is located between the motor **20** and the motor housing member connection passage **13b** in the radial direction of the rotary shaft **50**.

(67) Accordingly, the connection passage **70** is formed by the first impeller housing member connection passage **101**, the second impeller housing member connection passage **102**, and the motor housing member connection passage **13b**. The connection passage **70** is formed by the first impeller housing member **10b**, the second impeller housing member **10c**, and the connection passage defining housing member **13**. In other words, the connection passage **70** is formed by sequentially overlapping the first impeller housing member **10b**, the motor housing member **10a**, and the second impeller housing member **10c** with each other in the axial direction of the rotary shaft **50**.

(68) The centrifugal compressor **100** includes a third seal member **83**. The third seal member **83** is located between the second plate **17** and the first plate **16**. The third seal member **83** is, for example, an O-ring. The third seal member **83** seals the section between the second plate **17** and the first plate **16**. In the centrifugal compressor **100**, the first impeller housing member **10b** and the motor housing member **10a** are coupled to each other, with the third seal member **83** in between. As a result, the first impeller housing member connection passage **101** is connected to the motor housing member connection passage **13b**.

(69) The centrifugal compressor **100** includes a fifth seal member **85** and an eighth seal member **88**. The fifth seal member **85** and the eighth seal member **88** are located between the motor housing member **10a** and the third plate **18**. The fifth seal member **85** is located between the second motor housing member **15** and the third plate **18**. The eighth seal member **88** is located between the first motor housing member **14** and the third plate **18**. The fifth seal member **85** and the eighth seal member **88** are, for example, O-rings. The fifth seal member **85** seals the space between the second motor housing member **15** and the third plate **18**. The eighth seal member **88** seals the space between the first motor housing member **14** and the third plate **18**. In other words, the fifth seal member **85** and the eighth seal member **88** seal the space between the motor housing member **10a** and the third plate **18**. In the centrifugal compressor **100**, the motor housing member **10a** and the second impeller housing member **10c** are coupled to each other, with the fifth seal member **85** and the eighth seal member **88** located in between. As a result, the second impeller housing member connection passage **102** is connected to the motor housing member connection passage **13b**.

(70) The centrifugal compressor **100** includes a first seal member **81**, a second seal member **82**, a fourth seal member **84**, a sixth seal member **86**, and a seventh seal member **87**. The first seal member **81** seals the space between the first compressor housing member **11** and the second plate **17**. The second seal member **82** seals the space between the third plate **18** and the second compressor housing member **12**. The fourth seal member **84** seals the space between the second motor housing member **15** and the first plate **16**. The sixth seal member **86** seals the space between the second impeller housing member base **12a** and the second impeller housing member cover **19**. The seventh seal member **87** seals the space between the first plate **16** and the first motor housing member **14**. The first seal member **81**, second seal member **82**, fourth seal member **84**, sixth seal member **86**, and seventh seal member **87** are, for example, O-rings.

Operation of Present Embodiment

(71) The air drawn into the first impeller chamber **36** through the first suction port **35** is delivered to the first diffuser passage **38** while being accelerated by rotation of the first impeller **61**, and is then pressurized by passing through the first diffuser passage **38**. The air that has passed through the first diffuser passage **38** is discharged to the first discharge chamber **37**. The air discharged to the first discharge chamber **37** is discharged to the first discharge port **39**. Accordingly, the first discharge port **39** discharges the air that has been compressed by the rotation of the first impeller **61**.

(72) The air discharged to the first discharge port **39** flows from the first discharge port **39** to the first impeller housing member connection passage **101**. The air that has flowed through the first impeller housing member connection passage **101** is introduced into the motor housing member connection passage **13b**. The air introduced into the motor housing member connection passage **13b** is cooled by the coolant flowing through the cooling medium passage **13a** as the air passes through the connection hole **15d** in the motor housing member connection passage **13b**. In other words, the motor **20** is cooled by the coolant flowing through the cooling medium passage **13a**, and the air flowing through the connection passage **70** is also cooled.

(73) The air that has passed through the motor housing member connection passage **13b** is introduced into the first axial passage **401** through the third suction port **401b**. In other words, the third suction port **401b** faces the motor housing member connection passage **13b**, drawing in air in the axial direction of the rotary shaft **50**. The air that passed through the first axial passage **401** is introduced into the radial passage **403** through the third discharge port **401c**. In other words, the third discharge port **401c** opens so as to discharge, toward the guide **19d**, the air that has been drawn in from the third suction port **401b**.

(74) The air introduced into the radial passage **403** flows smoothly along the first cover connection surface **191**. In other words, the guide **19d** guides, in the radial direction of the rotary shaft **50**, the air that has been drawn in from the third suction port **401b** in the axial direction of the rotary shaft **50** such that the air approaches the second suction port **40**. Specifically, when air flows from the first axial passage **401** into the radial passage **403**, the air flows along the first cover connection surface **191**, thereby changing the air flow direction from the axial direction of the first axial passage **401** to the extending direction of the radial passage **403**. Accordingly, the first cover connection surface **191** changes the air flow direction from along the first axial passage defining surface **401a** to along the bottom surface **193**. The second impeller housing member cover **19** guides the air flowing from the first axial passage **401** to the radial passage **403** with the first cover connection surface **191**. This prevents sudden changes in the air flow direction. Thus, the second impeller housing member cover **19** prevents vortex formation when the air flow direction changes from the axial direction of the first axial passage **401** to the extending direction of the radial passage **403**. In this manner, the air flowing from the first axial passage **401** into the radial passage **403** is prevented from forming vortices.

(75) The air that has been introduced into the radial passage **403** passes through the radial passage **403** and is then introduced into the second axial passage **402**. The air flowing through the radial passage **403** moves smoothly along the second cover connection surface **192**. In other words, the air flowing in the radial direction of the rotary shaft **50** is further guided by the guide **19d** in the axial direction of the rotary shaft **50** toward the second suction port **40**. Specifically, when air flows from the radial passage **403** into the second axial passage **402**, the air flows along the second cover connection surface **192**. As a result, the air flow direction changes from the extending direction of the radial passage **403** to the axial direction of the second axial passage **402**. Accordingly, the second cover connection surface **192** changes the air flow direction from along the bottom surface **193** to along the second axial passage **402**. The second impeller housing member cover **19** guides the air flowing from the radial passage **403** into the second axial passage **402** with the second cover connection surface **192**. This prevents sudden changes in the air flow direction. Thus, the second impeller housing member cover **19** prevents vortex formation when the air flow direction changes

from the extending direction of the radial passage **403** to the axial direction of the second axial passage **402**. In this manner, the air flowing from the radial passage **403** toward the second suction port **40** is prevented from forming vortices.

(76) As described above, the third discharge port **401c** is connected to the second suction port **40** by the guide **19d**. In other words, the groove-shaped guide **19d**, which opens such that air flows from the third discharge port **401c** toward the second suction port **40**, is formed on the second impeller housing member cover **19**.

(77) The air that has passed through the second axial passage **402** is drawn into the second impeller chamber **41** through the second suction port **40**. The air drawn into the second impeller chamber **41** is delivered to the second diffuser passage **43** while being accelerated by rotation of the second impeller **62**, and is then pressurized by passing through the second diffuser passage **43**. The air that has passed through the second diffuser passage **43** is discharged to the second discharge chamber **42**. The air discharged to the second discharge chamber **42** is discharged to the second discharge port **44**. Accordingly, the second discharge port **44** discharges the air that has been compressed by the rotation of the second impeller **62**. The air discharged to the second discharge port **44** is supplied to the fuel cell stack **46** through the supply pipe **45**. The oxygen contained in the air supplied to the fuel cell stack **46** contributes to power generation in the fuel cell stack **46**.

Advantages of Present Embodiment

(78) The advantages of the present embodiment will now be described.

(79) (1) The connection passage **70** is formed by the first impeller housing member **10b**, the motor housing member **10a**, and the second impeller housing member **10c**, all of which are included in the housing **10**. In other words, the connection passage **70** is formed inside the housing **10**. For example, this prevents the formation of a gap between the housing **10** and a pipe, which is separate from the housing **10**, like in a configuration in which the first discharge port **39** and the second suction port **40** are connected to each other by attaching the pipe to the housing **10**. Thus, the centrifugal compressor **100** is prevented from being unnecessarily enlarged by an amount corresponding to the gap between the housing **10** and the pipe. Further, the connection passage **70** can be formed only by sequentially overlapping and coupling the first impeller housing member **10b**, the motor housing member **10a**, and the second impeller housing member **10c** with each other in the axial direction of the rotary shaft **50**. This eliminates, for example, the need to provide a structure in the pipe that can absorb the dimensional tolerances that occur between the housing **10** and the pipe when attaching the pipe to the housing **10**. Thus, the production efficiency of the centrifugal compressor **100** is improved. This improves the production efficiency of the centrifugal compressor **100** while also limiting its increase in size.

(80) (2) The cooling medium passage **13a** is located between the motor **20** and the motor housing member connection passage **13b** in the radial direction of the rotary shaft **50**. This allows the coolant flowing through the cooling medium passage **13a** to cool not only the motor **20** but also the air flowing through the motor housing member connection passage **13b**. Thus, the air cooled by the coolant can be drawn into the second impeller chamber **41** through the motor housing member connection passage **13b**, the second impeller housing member connection passage **102**, and the second suction port **40**. The air cooled by the coolant has a lower density than air that has not been cooled. In other words, when air is cooled before being drawn into the second impeller chamber **41**, the mass of the air being drawn into the second impeller chamber **41** increases. Thus, the second impeller **62** compresses a larger mass of air compared to when compressing air that has not been cooled. This improves the efficiency of air compression accompanying the rotation of the second impeller **62**.

(81) (3) For example, when constructing the cooling medium passage **13a** with a single housing member, it is necessary to manufacture such a housing member using a casting method that uses a core. The manufacturing method that uses a core incurs higher production costs than a manufacturing method that does not use a core. To solve this problem, the cooling medium passage

13a is defined such that coolant flows between the first motor housing member **14** and the second motor housing member **15**. Each of the first motor housing member **14** and the second motor housing member **15**, which define the cooling medium passage **13a**, does not need to be manufactured using the casting method that uses a core. In other words, in the motor housing member **10a**, which includes the first motor housing member **14** and the second motor housing member **15**, the cooling medium passage **13a** can be formed using the manufacturing method that does not use a core. As a result, the above configuration enables the formation of the cooling medium passage **13a** in the motor housing member **10a** while reducing manufacturing costs.

(82) (4) The third suction port **401b** draws, in the axial direction of the rotary shaft **50**, air that has flowed through the motor housing member connection passage **13b**. The guide **19d** guides, in the radial direction of the rotary shaft **50**, the air that has been drawn in from the third suction port **401b** in the axial direction of the rotary shaft **50** such that the air approaches the second suction port **40**. Further, the air flowing in the radial direction of the rotary shaft **50** is guided by the guide **19d** in the axial direction of the rotary shaft **50** toward the second suction port **40**. Thus, the guiding of air by the guide **19d** prevents the formation of vortices when the air flow direction changes. As a result, the second impeller housing member **10c** having the guide **19d** reduces the pressure loss of air flowing from the third suction port **401b** toward the second suction port **40**. Accordingly, the efficiency of air compression accompanying the rotation of the second impeller **62** is improved.

(83) (5) For example, there may be a case in which a single housing member forms a passage through which air flows from the third suction port **401b** to the second suction port **40**. The second suction port **40** extends in the axial direction of the rotary shaft **50** from the second impeller chamber **41**. At a position outward from the second suction port **40** in the radial direction of the rotary shaft **50**, the passage from the third suction port **401b** to the third discharge port **401c** extends in the axial direction of the rotary shaft **50** from the motor housing member connection passage **13b**. In other words, the passage through which air flows from the third discharge port **401c** to the second suction port **40** extends in the radial direction of the rotary shaft **50**. In order to manufacture such a housing member including the passage from the third suction port **401b** to the third discharge port **401c** and the passage from the third discharge port **401c** to the second suction port **40**, a casting method that uses a core may be employed. This leads to higher manufacturing costs.

(84) To solve such a problem, the second impeller housing member **10c** includes the second impeller housing member base **12a** and the second impeller housing member cover **19**. The second impeller housing member base **12a** includes the second suction port **40** and the passage that connects the third suction port **401b** to the third discharge port **401c**. The second impeller housing member cover **19** is connected to one of the surfaces of the second impeller housing member base **12a** where the third discharge port **401c** opens. Thus, the second impeller housing member cover **19** seals the opening at one end of the second suction port **40** and the opening defined by the third discharge port **401c**. Additionally, the passage between the second impeller housing member cover **19** and the second impeller housing member base **12a**, which is defined by the guide **19d** and the second impeller housing member base **12a**, is formed to connect the third discharge port **401c** and the second suction port **40** to each other. This eliminates the need for manufacturing with a casting method that uses a core, like in the case in which a single housing member forms a passage through which air flows from the third suction port **401b** to the second suction port **40**. Accordingly, increases in the manufacturing costs are avoided.

(85) (6) The inverter **65** is provided on the housing outer surface **10e**. That is, the coolant flowing through the cooling medium passage **13a** cools the motor **20**, the air flowing through the connection passage **70**, and the inverter **65**. In other words, the configuration of forming the connection passage **70** inside the housing **10** and placing the inverter **65** on the outer surface of the housing **10** reduces the size of the centrifugal compressor **100**. Further, a single cooling medium passage **13a** can simultaneously cool the motor **20**, the air flowing through the connection passage

70, and the inverter **65**. This improves the efficiency of compression by the second impeller **62** and the efficiency of dissipating the heat generated during operation of the centrifugal compressor **100**, while limiting an increase in the size of the centrifugal compressor **100**.

(86) (7) Multiple annular fins **14d** extend along the outer circumferential surface of the circumferential wall **14b** on the outer circumferential surface of the first motor housing member **14**. Thus, coolant flowing through the cooling medium passage **13a** flows along each fin **14d** in the circumferential direction of the first motor housing member **14** and the second motor housing member **15**. This allows for efficient heat exchange with the first motor housing member **14**, second motor housing member **15**, and inverter housing member **66**. In other words, coolant flows along each fin **14d** so as to efficiently cool the motor **20**, the air flowing through the cooling medium passage **13a**, and the inverter **65**. Here, for example, the configuration in which the annular fins **14d** extending along the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14** are protruded into the cooling medium passage **13a** and the connection passage **70** is formed in the housing **10**, may be achieved by a single housing member. Manufacturing such a housing member requires the use of a casting method that uses a core, leading to higher manufacturing costs. In the present embodiment, the first motor housing member **14** is separate from the second motor housing member **15**. In such a configuration, each of the first motor housing member **14** and the second motor housing member **15** does not need to be manufactured using the casting method that uses a core. This allows for, while reducing manufacturing costs, the configuration in which the annular fins **14d** extending along the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14** are protruded into the cooling medium passage **13a** and the connection passage **70** is formed in the housing **10**.

Modifications

(87) The above embodiment may be modified as follows. The above embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

(88) The facing surface **19b** of the impeller housing member cover **19** does not have to include the guide **19d**. In this case, for example, a recess extending in the radial direction of the rotary shaft **50** is formed on the axial end face **12d** of the second impeller housing member base **12a**, and this recess may function as the guide **19d**. That is, the second impeller housing member base **12a** may include the guide **19d**.

(89) The second impeller housing member **10c** does not have to include the second impeller housing member cover **19**. In this case, the second impeller housing member **10c** includes the second impeller housing member base **12a** and the third plate **18**. In this case, the first axial passage **401** and the second suction port **40** are connected to each other through the second impeller housing member base **12a** without opening in the axial end face **12d**. That is, in this case, the second impeller housing member **10c** does not need to include the third discharge port **401c**.

(90) The second impeller housing member base **12a** does not have to include the second axial passage **402**. In this case, the second suction port **40** is directly connected to the radial passage **403**.

(91) The motor housing member **10a** does not have to include the first motor housing member **14** and the second motor housing member **15**. In other words, the motor housing member **10a** may include a single housing component. In this case, the cooling medium passage **13a** is formed inside the housing **10**.

(92) The cooling medium passage **13a** does not have to be provided between the motor **20** and the motor housing member connection passage **13b** in the radial direction of the rotary shaft **50**.

(93) The centrifugal compressor **100** does not have to include the cooling medium passage **13a**.

(94) The centrifugal compressor **100** may have a configuration in which a single thin fin extends helically along the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14**.

(95) Fins do not have to be provided on the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14**. In this case, the coolant flowing through the cooling medium passage **13a** may flow along the outer circumferential surface of the circumferential wall **14b** of the first motor housing member **14** in the axial direction of the circumferential wall **14b**.

(96) The cooling medium flowing through the cooling medium passage **13a** may be a substance other than coolant. For example, the cooling medium may be refrigerant or air.

(97) The inverter **65** does not have to be provided on the housing outer surface **10e**.

(98) The fluid that is to be compressed by the centrifugal compressor **100** is not limited to air. Thus, the centrifugal compressor **100** may be employed in any suitable application to compress any type of fluid. For example, the centrifugal compressor **100** may be employed in an air conditioner to compress refrigerant, which is fluid. Further, the centrifugal compressor **100** may be mounted on any structure other than a vehicle.

(99) Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. A centrifugal compressor, comprising: a rotary shaft; a motor that rotates the rotary shaft; a first impeller that rotates integrally with the rotary shaft to compress a fluid; a second impeller located on an opposite side of the rotary shaft from the first impeller, wherein the motor is arranged between the first impeller and the second impeller, and the second impeller rotates integrally with the rotary shaft; a housing including: a first impeller housing member that defines a first impeller chamber accommodating the first impeller, the first impeller housing member including a first suction port that draws the fluid into the first impeller chamber and a first discharge port that discharges the fluid compressed by rotation of the first impeller; a motor housing member that defines a motor chamber accommodating the motor; and a second impeller housing member that defines a second impeller chamber accommodating the second impeller, the second impeller housing member including a second suction port that draws the fluid into the second impeller chamber and a second discharge port that discharges the fluid compressed by rotation of the second impeller; and a connection passage that connects the first discharge port to the second suction port such that the fluid flows from the first discharge port toward the second suction port, wherein the connection passage includes: a first impeller housing member connection passage located in the first impeller housing member and connected to the first discharge port; a second impeller housing member connection passage located in the second impeller housing member and connected to the second suction port; and a motor housing member connection passage extending through the motor housing member and connecting the first impeller housing member connection passage to the second impeller housing member connection passage, and the connection passage is formed by sequentially overlapping the first impeller housing member, the motor housing member, and the second impeller housing member with each other in an axial direction of the rotary shaft.

2. The centrifugal compressor according to claim 1, wherein the motor housing member includes a cooling medium passage through which a cooling medium flows to cool the motor, and the cooling medium passage is located between the motor and the motor housing member connection passage

in a radial direction of the rotary shaft.

3. The centrifugal compressor according to claim 2, wherein the motor includes: a cylindrical stator; and a rotor provided on the rotary shaft, the rotor being configured to rotate integrally with the rotary shaft, the motor housing member includes: a first motor housing member having an inner circumferential surface to which the stator is fixed; and a second motor housing member through which the motor housing member connection passage extends, and the cooling medium passage is defined such that the cooling medium flows between the first motor housing member and the second motor housing member.

4. The centrifugal compressor according to claim 1, wherein the second impeller housing member includes: a third suction port that faces the motor housing member connection passage to draw in the fluid in the axial direction; and a guide that guides, in a radial direction of the rotary shaft, the fluid drawn in from the third suction port in the axial direction such that the fluid approaches the second suction port, and further guides, toward the second suction port in the axial direction, the fluid flowing in the radial direction.

5. The centrifugal compressor according to claim 4, wherein the second impeller housing member includes: a second impeller housing member base in which the third suction port, the second suction port, and the second discharge port open and in which a third discharge port opens such that the fluid drawn in from the third suction port is discharged toward the guide; and a second impeller housing member cover that connects the third discharge port to the second suction port, the second impeller housing member cover including the guide opening such that the fluid flows from the third discharge port toward the second suction port, and the guide having a groove shape.
